Rhythm, Sound & the Science of Music

Science • Grades 5–8

GRADES 5–8 | 30 MINUTES | SCIENCE

LEARNING OBJECTIVE

What Students Will Understand

Students will understand the physics of sound — frequency, amplitude, and resonance — through the lens of jazz and blues instruments. They will connect the science of how instruments produce sound to why different instruments have distinct timbres, and explore how the human ear and brain process music.

STANDARDS ALIGNMENT

NGSS MS-PS4-1	Use mathematical representations to describe waves — including wave frequency, amplitude, and wavelength
NGSS MS-PS4-2	Develop and use a model to describe how waves transfer energy
CCSS.ELA-LITERACY RI.6.7	Integrate information presented in different media formats including visual, quantitative, and oral

MATERIALS NEEDED

- Device to play audio clips of individual instruments (trumpet, saxophone, double bass, piano, drums)
- Rulers or string for simple wave demonstrations
- Differentiated student handouts
- Optional: tuning fork to demonstrate vibration

HOOK (5 MINUTES)

Hold up a tuning fork (or tap a glass). Ask: 'What is sound, actually? What's physically happening when you hear music?' Take responses. Then play 10 seconds each of a trumpet, a bass, and a piano from the Jazz Lab — ask students what's physically different between them.

DIRECT INSTRUCTION (10 MINUTES)

"Every sound you hear is a wave — a vibration moving through air. High notes vibrate faster (high frequency). Low notes vibrate slower (low frequency). Loud sounds have bigger waves (high amplitude). Soft sounds have smaller waves. When Duke Ellington's band played in Harlem in 1927, every instrument in that room was creating its own unique wave pattern — and your ears were doing the extraordinary work of separating them all out."

"The timbre — the unique colour of each instrument — comes from its overtones. A trumpet playing middle C and a piano playing middle C produce the same fundamental frequency. But the trumpet has different overtone patterns from the piano, which is why they sound completely different. Jazz composers like Ellington understood this intuitively — he wrote specific parts for specific instruments because he knew exactly what wave patterns each one produced."

ACTIVITY (15 MINUTES)

Differentiated Work Time

Distribute student handouts. Students self-select their challenge level. Circulate with targeted questions to push thinking at each level.

🟢 Green Level

Scaffolded activities with vocabulary support and structured response frames. Students engage with core content with guidance.

🟡 Yellow Level

Independent grade-level analysis. Students apply concepts, make connections, and express their own thinking without heavy scaffolding.

🔴 Red Level

Extended critical thinking and research. Students synthesise across sources, evaluate arguments, and produce original analysis or creative work.

Note: Red level activities may extend beyond the lesson period — they are designed to continue at home.

CLOSING (5 MINUTES)

Bring the class back together. Ask 2–3 students from different levels to share one thing they discovered or created. Close with:

> *"The Harlem Renaissance wasn't just a historical event. It was a demonstration that ordinary people — with talent, community, and courage — can create something that changes the world. The work you've done today connects you to that tradition."*

ASSESSMENT

Formative: Circulate during work time and check for understanding of the core concepts. Use targeted questions to assess depth of thinking.

Exit Ticket: "What is one connection between the Harlem Renaissance and Science?"

🏛 Mini Museum Connection

Option 1 — Visit the Exhibition: If the Harlem Renaissance exhibition is available at your school, visit the Mini Museum to experience the phonograph, Jazz Lab, Great Migration Map, and more. Ask your librarian to arrange a class visit.

Option 2 — Virtual Visit: Explore the full exhibition at **minimuseumproject.com/exhibitions/harlem-renaissance**

